AJAY SRIVATSAN R.



CONTACT

Phone: +91 7418694168
Email: ajju2004srivatsan@gmail.com
GitHub: github.com/ajju2004srivatsan-cloud

ACADEMIC HISTORY

Sakthi Vinayakar Hindu Vidyalaya, Tuticorin
High School Education

- 10th Boards | 2020 - **85.0%**
- 12th Boards | 2022 - **86.6%**

Amrita Vishwa Vidyapeetham, Coimbatore
Bachelor of Technology in Electronics and Communication Engineering

- CGPA - **7.98**
- Oct, 2022 - Jul, 2026

WORK EXPERIENCE

- **Apsis Solutions — Embedded Systems & Hardware Design Intern | March 2026 – May 2026**

  Developed and debugged embedded firmware for drone and defence applications while contributing to commercial products through schematic review, debugging, and prototyping hardware.

- **OEG Defence & Aerospace Pvt. Ltd. — Software Engineer | June 2026 – Present**

  Developing embedded software and communication systems for UAV, aerospace, and defence platforms. Working on ArduPilot integration, MAVLink, RF communication, networking, embedded Linux, software validation, and hardware-software integration for autonomous systems.

TECHNICAL SKILLS

- **Programming Languages:** Python, C, C++, Assembly, Java, HTML
- **AI & ML:** TensorFlow Lite, PyTorch, OpenCV, Librosa, MCP
- **Communication:** SDR, SDN, FHSS, Wireless Sensor Networks, Spread Spectrum
- **Hardware & Systems:** HFSS, Docker, VHDL, Linux, ROS2

SOFTWARE AND TOOLS EXPERTISE

- **Design & Simulation:** LTspice, WinSpice, Ansys HFSS, Gazebo
- **Machine Learning:** TensorFlow, PyTorch, Edge Impulse
- **Embedded Systems:** Keil µVision4, STM32CubeProgrammer, GNU Radio
- **Networking & Emulation:** GNU Radio, Mininet, Wireshark, OpenFlow, POX Controller

PROJECTS

- **Clustered Multi-UAV Swarm Command System — Embedded Systems & Network Architecture**

  Engineered an adaptive, clustered UAV command system for Indian Army, successfully transitioned from a technical demonstration to active operational use by North Eastern Cavalry units. Hardwired directly to flight controllers, eliminating the need for intermediate interfaces.

- **Vision-Based Autonomous Drone Navigation System — Computer Vision, VIO & Robotics**

  Built a GNSS-denied navigation pipeline fusing Visual-Inertial Odometry, optical flow, and IMU data. Implemented satellite-image localization to correct accumulated drift, improving position accuracy. Integrated the stack with flight controllers via MAVLink for real-time autonomous control using Jetson.

- **Lightweight Audio-Visual Speech Enhancement System — Deep Learning & Multimodal AI**

  Designed a multimodal pipeline combining CRNN-based audio processing with ResNet lip-feature extraction to enhance speech in noisy environments. Reduced model size for real-time use cases while improving PESQ, STOI, and SNR audio-only baselines. It was accepted for presentation at NGNDAI, MNNIT Allahabad (2026).

- **Autonomous Robotics Simulation — Simulation & Robotics Integration**

  Developed and tested autonomous systems using Gazebo and NVIDIA Isaac Sim, covering multi-UAV swarm coordination, computer vision deployment, and robotic system simulations. Worked on software-in-the-loop testing.

- **Real Estate ERP Platform — React, PostgreSQL & Linux VPS**

  Deployed a full-stack ERP system for a real estate company, used daily by 30+ active users across sales, accounts, site management, and HR teams. Handled concurrent usage with multi-threaded server setup, keeping average response time under 10 ms.

CONFERENCES & WORKSHOPS

- **SoCTA 2025 – International Conference on Soft Computing: Theories and Applications**

  NIT Jalandhar* | *Presentation: 2025* | *Publication: 2026

  Presented *"A Comparative Study of Trapdoor Sampling Algorithms for Lattice-Based Cryptography"* at SoCTA 2025; publication scheduled for 2026.

- **Building Edge AI Application on the MAX 7800X | Analog Devices**

  NIT Trichy* | *March 2024

  The workshop hosted by NIT Trichy (March 2024) focused on learning the MAX 7800X microcontroller to develop efficient Edge AI solutions.

- **Digital Design Mastery with FPGA | Vivado System Design**

  Amrita School of Engineering, Coimbatore* | *September 2024

  A workshop organized by Amrita School of Engineering, Coimbatore, in September 2024 offered a hands-on experience to explore FPGA-based digital design techniques using Vivado.